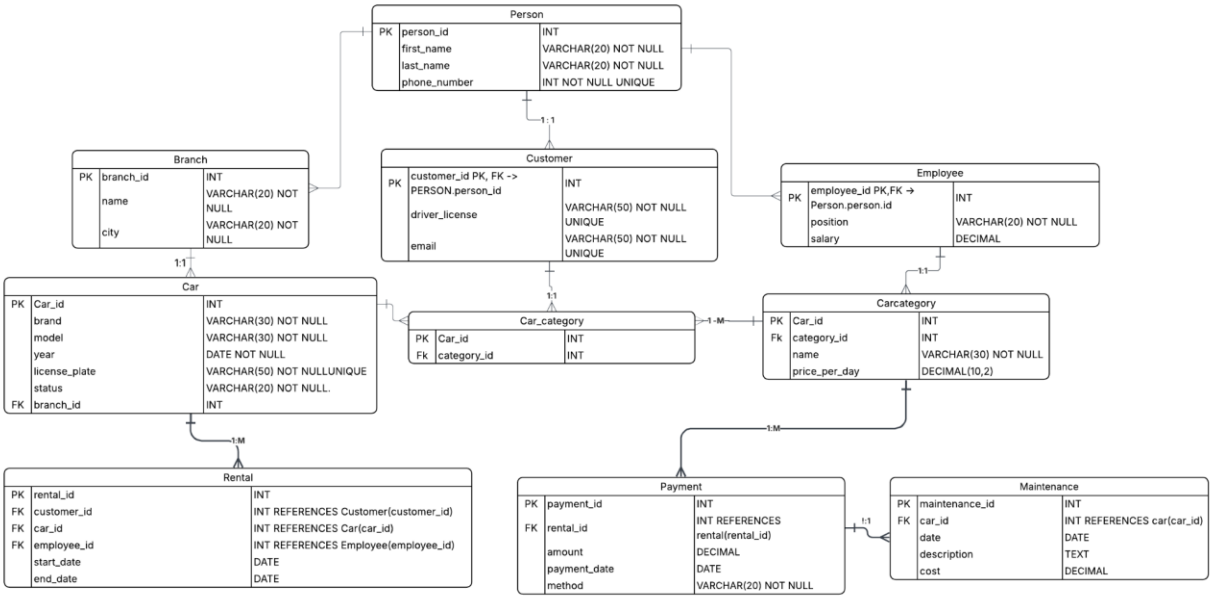
1.1 Logical Scheme



1.2 Objects

Table: Person

Stores personal information for all individuals in the system — both customers and employees. Each person has a unique identifier (person_id) and is identified by their first and last name and a unique phone number used for contact purposes.

Table Description

Field Name	Data Type / Constraints	Description
person_id	INT, PK	Unique identifier for each person. Auto-incremented primary key.
first_name	VARCHAR(20), NOT NULL	Given name of the person. Cannot be empty.
last_name	VARCHAR(20), NOT NULL	Family name of the person. Cannot be empty.
phone_number	INT, NOT NULL, UNIQUE	Contact phone number. Must be unique across all persons.

Example Data

person_id	first_name	last_name	phone_number
1	Alexei	Morozov	87011234567
2	Dinara	Bekova	87029876543
3	Carlos	Fernandez	87035551234

person_id	first_name	last_name	phone_number
4	Fatima	Al-Rashid	87040001122

Table: Customer

Extends the Person entity with rental-specific customer information. A customer must have a valid driver's license to rent a vehicle. Each customer record is linked to a Person record via customer_id (FK). Email addresses are stored for billing notifications and account management.

Table Description

Field Name	Data Type / Constraints	Description
customer_id	INT, PK/FK	References person_id from Person table. Identifies the customer.
driver_license	VARCHAR(50), UNIQUE	Official driver's license number. Must be unique; required for vehicle rental.
email	VARCHAR(50), UNIQUE	Customer email address used for booking confirmations and receipts.

Example Data

customer_id	driver_license	email
1	KZ1234567	alexei.morozov@mail.ru
2	KZ7654321	dinara.bekova@gmail.com
3	US9876543	carlos.fern@outlook.com

Table: Employee

Contains employment details for staff members of the car rental company. Each employee is also a Person and shares a common identifier. This table tracks the employee's role within the company and their monthly salary in decimal format for precise financial calculations.

Table Description

Field Name	Data Type / Constraints	Description
employee_id	INT, PK/FK	References person_id from Person table. Identifies the employee.
position	VARCHAR(20)	Job title or role within the company (e.g., Manager, Clerk, Technician).
salary	DECIMAL	Monthly gross salary of the employee in local currency (KZT equivalent).

Example Data

employee_id	position	salary
4	Manager	500000.00
5	Clerk	250000.00
6	Technician	300000.00

Table: Branch

Represents physical office locations of the car rental company. Each branch is identified by a unique ID and has a display name and city. Branches serve as hubs where vehicles are stored, rented, and maintained. Cars and employees are assigned to branches.

Table Description

Field Name	Data Type / Constraints	Description
branch_id	INT, PK	Unique identifier for each branch location.
name	VARCHAR(20)	Human-readable branch name (e.g., 'Central', 'West Branch').
city	VARCHAR(20)	City where the branch is located. Used for geographic filtering.

Example Data

branch_id	name	city
1	Central Office	Atyrau
2	West Branch	Aktau
3	North Branch	Oral

Table: Car

Stores all vehicles available in the rental fleet. Each car has a unique license plate and is linked to a specific branch. The status field tracks real-time vehicle availability. This table is central to the system as it connects branches, categories, rentals, and maintenance records.

Table Description

Field Name	Data Type / Constraints	Description
car_id	INT, PK	Unique identifier for each vehicle in the fleet.
brand	VARCHAR(30)	Manufacturer brand of the car (e.g., Toyota, BMW, Mercedes).
model	VARCHAR(30)	Specific model name of the vehicle (e.g., Camry, X5).
year	DATE	Manufacturing year of the vehicle. Affects pricing tier and depreciation.

Field Name	Data Type / Constraints	Description
license_plate	VARCHAR(50), UNIQUE	Official vehicle registration number. Must be unique across all cars.
status	VARCHAR(20)	Current status: 'Available', 'Rented', or 'Maintenance'.
branch_id	INT, FK	References Branch table. Indicates which branch currently holds the car.

Example Data

car_id	brand	model	year	license_plate	status	branch_id
1	Toyota	Camry	2020	123ABC01	Available	1
2	BMW	X5	2021	456DEF02	Rented	1
3	Hyundai	Elantra	2019	789GHI03	Maintenance	2
4	Kia	Sportage	2022	321JKL04	Available	3
5	Mercedes	C200	2020	654MNO05	Rented	2

Table: Category

Defines vehicle categories used to classify cars and set daily rental pricing. Categories allow customers to choose vehicles by type and budget. A vehicle can belong to multiple categories through the Car_Category junction table.

Table Description

Field Name	Data Type / Constraints	Description
category_id	INT, PK	Unique identifier for each category.
name	VARCHAR(30)	Category label such as Economy, SUV, or Luxury.
price_per_day	DECIMAL(10,2)	Daily rental rate in USD for vehicles under this category.

Example Data

category_id	name	price_per_day
1	Economy	30.00
2	SUV	60.00
3	Luxury	120.00

Table: Car_Category

A junction (associative) table that implements a many-to-many relationship between cars and categories. A single vehicle can be listed under multiple categories, and a category can contain many vehicles. Both fields together form the composite primary key.

Table Description

Field Name	Data Type / Constraints	Description
car_id	INT, PK/FK	References Car table. Part of the composite primary key.
category_id	INT, FK	References Category table. Links the car to its category.

Example Data

car_id	category_id
1	1
2	3
3	1
4	2

Table: Rental

Records each car rental transaction in the system. A rental links a customer to a specific vehicle and the employee who processed the rental. Start and end dates define the rental period, which determines the total amount charged via the Payment table.

Table Description

Field Name	Data Type / Constraints	Description
rental_id	INT, PK	Unique identifier for each rental transaction.
customer_id	INT, FK	References Customer. Identifies who rented the vehicle.
car_id	INT, FK	References Car. Specifies which vehicle was rented.
employee_id	INT, FK	References Employee. Staff member who processed the rental.
start_date	DATE	Date the rental period begins. Used to calculate charges.
end_date	DATE	Date the rental period ends. Car must be returned by this date.

Example Data

rental_id	customer_id	car_id	employee_id	start_date	end_date
1	1	2	4	2026-04-01	2026-04-05
2	2	4	5	2026-04-02	2026-04-06
3	3	5	4	2026-04-03	2026-04-07

rental_id	customer_id	car_id	employee_id	start_date	end_date
4	1	6	6	2026-04-04	2026-04-08

Table: Payment

Tracks all payments made by customers for their rentals. Each payment is linked to a specific rental and records the total amount charged, the date of payment, and the method used. Multiple payment methods are supported including card, cash, and online transfer.

Table Description

Field Name	Data Type / Constraints	Description
payment_id	INT, PK	Unique identifier for each payment record.
rental_id	INT, FK	References Rental. Links the payment to its corresponding rental.
amount	DECIMAL	Total payment amount in USD, based on rental duration and category rate.
payment_date	DATE	Date the payment was processed and confirmed.
method	VARCHAR(20)	Payment method used: 'Card', 'Cash', or 'Online'.

Example Data

payment_id	rental_id	amount	payment_date	method
1	1	200.00	2026-04-01	Card
2	2	240.00	2026-04-02	Cash
3	3	500.00	2026-04-03	Card
4	4	300.00	2026-04-04	Online

Table: Maintenance

Logs all maintenance and repair events performed on fleet vehicles. Each record captures the car involved, the date of service, a description of the work done, and the associated cost. This helps track vehicle condition over time and plan preventive maintenance schedules.

Table Description

Field Name	Data Type / Constraints	Description
maintenance_id	INT, PK	Unique identifier for each maintenance event.
car_id	INT, FK	References Car table. Identifies which vehicle was serviced.
date	DATE	Date when maintenance was performed.
description	TEXT	Detailed description of the maintenance work done (e.g., oil change, tire replacement).

Field Name	Data Type / Constraints	Description
cost	DECIMAL	Total cost of the maintenance service in USD.

Example Data

maintenance_id	car_id	date	description	cost
1	3	2026-03-01	Full oil change and filter replacement	50.00
2	7	2026-03-05	All four tires replaced due to wear	200.00
3	2	2026-03-10	Brake pads inspected and replaced	100.00
4	5	2026-03-15	Engine diagnostics and spark plug change	150.00
5	1	2026-03-20	Battery replacement and electrical check	120.00